

CampusOne — Master Specification & Prompt Engineering Book

Nitte Meenakshi Institute of Technology (NMIT) Bengaluru | Databricks Genie Hackathon 2026

1. Executive Summary & Problem Statement

CampusOne is the definitive AI-native student intelligence platform engineered specifically for **Nitte Meenakshi Institute of Technology (NMIT)**, Bengaluru. Built on top of **Databricks Free Edition** and driven by a multi-mode **Databricks Genie Agent** over Unity Catalog governed Delta tables, CampusOne solves the chronic campus discovery barrier faced by students in crisis: finding project collaborators across 10 engineering departments, navigating SIH hackathon team assembly, discovering co-curricular clubs, and seeking data-backed academic guidance.

Theme Alignment: Genie-Powered Campus Intelligence (Bengaluru Tech Week / Databricks Hackathon 2026). CampusOne combines real institutional facts (NMIT official alumni, verified clubs, placement stats) with 80+ synthetic student profiles to deliver a data-informed 'What-If?' simulator, natural language team matching, and source-linked campus news.

2. NanoBanana Logo Generation Prompt

Use the exact prompt below in **NanoBanana / Midjourney / DALL-E 3** to generate the official minimalist logo icon for CampusOne:

Minimalist vector logo for 'CampusOne', an AI-native university intelligence platform for Nitte Meenakshi Institute of Technology (NMIT). Abstract geometric combination of a stylized letter 'C' intertwined with a single solid numeral '1', integrated with a subtle glowing neural node network spark. Colors: Deep Navy Blue (#1E3A8A), Verified Teal (#0F766E), and Charcoal (#0B0B0C) on a clean, solid pure white background (#FFFFFF). Flat vector design, modern SaaS app icon style, zero gradients, 8k resolution, crisp vector contours, ultra-clean UI branding --no 3d, realistic photos, mockups, shadows.

3. Feature Architecture & UI Placement Map

Feature Name	Core Value & Description	UI Location
F1: Mandatory Onboarding & Identity	Simple OTP authentication restricted to @nitte.edu.in or @nmit.ac.in. Multi-step wizard capturing skills, interests, goals, and granular field-level privacy controls.	/login & /onboarding
F2: Find My People (NL Search)	Natural language semantic search powered by v_people_search view. Returns student cards with transparent WhyMatch chips (e.g. 'Matching skills: PyTorch + Shared interest: Drones').	/people
F3: Opportunities & Team Assembly	Hackathons (SIH 2026), research assistantships, and internships with Genie Fit Scores. Includes 'Build My Team' constraint solver assembling 3–5 balanced student teams with skill coverage checklist and gap callout.	/opportunities & /opportunities/:id
F4: Clubs & Culture Wall	Co-curricular directory (NMIT Hacks, GDG on Campus, E-Cell NMIT, etc.) with verified Instagram/website links, photo post feeds, recruitment badges, and personalized 'Good for you if...' recommendation chips.	/clubs & /clubs/:id
F5: Campus Digest & Alumni Stories	Editorial news feed in Source Serif 4 typography featuring real NMIT announcements and 10 verified alumni (Kathmandu Mayor Balen Shah, UK Space Agency Lead Dr. Mamatha, etc.) with direct nitte.edu.in source links.	/stories
F6–F8: Universal Genie Agent	Multi-mode conversational AI (General Q&A; / Academic Guidance / What-If Simulator). Features prompt-injection defense, structured card rendering, and mandatory estimate disclaimer footnotes.	/genie
F9–F10: Scannable Home & Feedback	7-tier scannable homepage with urgent action banner, capped recommendation strips (max 3 items), and More / Less like this feedback affordances.	/ (Home)

4. Implementation Plan & Prompts in 10 Batches

Batch 0 — Foundation & Repository Scaffolding

Scaffold React+TS+Vite frontend and FastAPI backend with strict ESLint, Prettier, folder structure, .env.example, and /health endpoint.

Executable Agent Prompt:
Create a production Vite+React+TS app in campusone/frontend with strict tsconfig, Inter font, and a FastAPI app in campusone/backend with CORS and /health endpoint.

Batch 1 — Design System & Shared Components

Build core design tokens in tokens.css (#FFFFFF, #0B0B0C, #1E3A8A, Inter, Source Serif 4) and 13 UI components (Button, Card, Avatar, Badge, Input, TextArea, TagInput, Modal, Tabs, NavBar, EmptyState, LoadingState, ErrorState).

Executable Agent Prompt:
Build reusable design tokens and UI components in frontend/src/components/ui ensuring 0 hardcoded hex codes outside tokens.css and clean mobile responsive collapse.

Batch 2 — Databricks Free Edition Data Layer & Seeding

Write schema.sql for Unity Catalog catalog campusone schema core with 17 Delta tables and 5 governed semantic views. Seed verified NMIT facts (10 alumni, 8 clubs) and 80 synthetic student profiles via seed_data.py.

Executable Agent Prompt:

Create schema.sql and seed_data.py creating Unity Catalog views v_people_search, v_opportunity_fit, v_club_culture, v_whatif_patterns, v_campus_digest with fallback SQLite support.

Batch 3 — CampusOne Identity, Auth & Privacy (F1)

Implement session auth for @nitte.edu.in emails, 7-step mandatory onboarding wizard, and server-side privacy enforcement popping usn_encrypted and email_hash for non-owners.

Executable Agent Prompt:

Build backend routers auth.py & profile.py and frontend pages Login.tsx, Onboarding.tsx, ProfileEdit.tsx, ProfileView.tsx with strict field-level privacy masking.

Batch 4 — Find My People & Team Assembly (F2 + F3)

Build /genie/people-search with natural language query parsing and transparent why_match chips. Build /opportunities/:id/build-team constraint solver proposing 3-5 distinct members with skill coverage checklist.

Executable Agent Prompt:

Implement genie_service.py, team_builder.py, /people search UI, OpportunityDetail.tsx team modal, and /connections API.

Batch 5 — Opportunities Hub & Fit Scorer (F3 continued)

Build /opportunities list endpoint with Genie fit calculation (65% skill overlap + 20% interest bonus + 15% dept relevance). Add deadline urgency visual treatment (< 72h warning badge) and verified vs synthetic badging.

Executable Agent Prompt:

Implement fit_scorer.py, opportunities.py router, OpportunitiesList.tsx with type filter pills, and OpportunityDetail.tsx with official nitte.edu.in link out.

Batch 6 — Clubs & Culture Wall (F4)

Build /clubs list with culture tags and personalized 'Good for you if...' lines. Build /clubs/:id with post feed, official Instagram/website links, and pinned /genie/ask-club/:id' Q&A; input box.

Executable Agent Prompt:

Implement club_service.py, clubs.py router, ClubsDirectory.tsx, and ClubDetail.tsx with anti-hallucination scoping for unverified member counts.

Batch 7 — Genie Centerpiece: Universal, Academic, What-If (F6+F7+F8)

Build multi-mode /genie/chat endpoint (General / Academic / What-If). Implement 3-vector prompt injection defense (stripping malicious control vectors), inline routing to People Search, and What-If comparison card with estimate disclaimer footnote.

Executable Agent Prompt:

Implement genie_engine.py, genie_chat.py router, and GenieChat.tsx with 3 mode tabs, example prompt chips, and conversation turn logging.

Batch 8 — Campus Pulse, Recommendations & Digest (F9+F10+F5)

Build /home aggregation endpoint (urgent item waterfall P1->P2->P3, top 3 people, top 3 opps, 1 pulse, 1 story). Build /feedback endpoint for more/less signals and /stories Campus Digest styled in Source Serif 4.

Executable Agent Prompt:

Implement home_service.py, home_digest.py router, scannable Home.tsx, and editorial Stories.tsx with nitte.edu.in source links.

Batch 9 — QA, Privacy & Hardening

Run automated test suite (run_full_qa.py) verifying happy path, empty states, contradiction resolution, privacy boundaries, stale source dates, synthetic badging, 3-vector prompt injections, and 10-question anti-hallucination benchmark.

Executable Agent Prompt:

Build run_full_qa.py and execute 11-category audit matrix to guarantee 100% test coverage and zero privacy leaks before deployment.

Batch 10 — Deployment & Demo Packaging

Create DEPLOY.md guide for 20-minute stack reproduction, literal 90-second DEMO_SCRIPT.md timeline, fallback recording archive, and final visual palette hex audit.

Executable Agent Prompt:

Package production bundle via npm run build, write DEPLOY.md and DEMO_SCRIPT.md, and push commit 'deployed campusone platform' to GitHub.